Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State what is meant by a member in tension.

- The member is being pulled / stretched by the forces at its ends

3 Quick check

- a) 3.8 sin 62
- b) Strut
- c) 980 N
- d) 14.94 kN at 4.15 m

4 Key terms

- 1 – C
- 2 – B
- 3 – D
- 4 – F
- 5 – E
- 6 – A

5 State the two conditions of equilibrium.

- The sum of the forces = 0 (upward = downward, left = right)
- The sum of the moments = 0 (clockwise = anticlockwise)

6 Calculate each reaction.

- Total load = 2 × 6 = 12 kN acting at the centre
- Each reaction = 6 kN

7 Calculate the magnitude and direction of the resultant.

- $R = \sqrt{(400^2 + 300^2)} = 500$ N
- $\theta = \tan^{-1}(300 \div 400) = 36.9°$
- Direction stated relative to the horizontal